Signorini's Problem

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Contents

Abstract

This is the abstract.[1]

1 Introduction

This is the Introduction.

2 Mathematical Framework

2.1 Sobolev Spaces and the Trace Operator

In this subsection, we will make a brief excursion into selected topics in functional analysis, providing the necessary theory for the subsequent subsections. In classical analysis, derivatives are defined pointwise and require a certain level of smoothness. However, many functions of interest fall outside this framework. To extend the concept of differentiation to a broader class of functions, we can weaken what it means to *have a derivative*. The key idea is to characterize derivatives not by their pointwise behavior, but by how they act under integration against *test functions*. Using integration by parts, one can express the derivative of a smooth function in a form that remains meaningful even when the function itself is not differentiable in the classical sense. This leads naturally to the concept of weak derivatives, which allow us to treat integrable functions as *differentiable* in a generalized sense. We start by defining two important function spaces.

Definition 2.1 (Locally Integrable Functions)**.** *The space $L^1_{\mathrm{loc}}(\Omega)$ of all* `locally integrable` *functions on Ω is defined as*

$$L^1_{\mathrm{loc}}(\Omega) := \left\{ f : \Omega \to \mathbb{R} \text{ measurable} \;\middle|\; f|_K \in L^1(K)\ \forall K \subset \Omega,\ K \text{ compact} \right\}.$$

Definition 2.2. *The space $\mathcal{C}_c^\infty(\Omega)$ contains all infinitely often differentiable functions with compact support in Ω, i.e.*

$$\mathcal{C}_c^\infty(\Omega) = \{ f \in \mathcal{C}^\infty(\Omega) \mid \mathrm{supp}(f) \subset \Omega \text{ compact} \}$$

and $u \in \mathcal{C}_c^\infty(\Omega)$ is called a `test function`*.*

With these functions in hand, we can now extend the classical concept of differentiability to the concept of *weak* differentiability.

Definition 2.3 (Weak Derivatives)**.** *Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and $\alpha \in \mathbb{N}_0^d$. A function $g \in L^1_{\mathrm{loc}}(\Omega)$ is called* `weak derivative` *of u of order α if*

$$\int_\Omega u\, \partial^\alpha \varphi \,\mathrm{d}x = (-1)^{|\alpha|} \int_\Omega g\, \varphi \,\mathrm{d}x \quad \forall \varphi \in \mathcal{C}_c^\infty(\Omega)$$

If such g exists, it is unique up to null sets and we write $\partial^\alpha u = g$ and keep the notation the same as in the strong sense of derivatives.

Of course, there are also spaces of weakly differentiable functions, named after the Soviet mathematician Sergei Lvovich Sobolev (1908–1989).

Definition 2.4 (Sobolev Spaces)**.** *For $k \in \mathbb{N}_0$ and $1 \le p \le \infty$ we define the* `Sobolev space` *$W^{k,p}(\Omega)$ to be the space of every L^p-function whose weak derivatives up to order k are also L^p-functions, i.e.*

$$W^{k,p}(\Omega) := \{ u \in L^p(\Omega) \mid \partial^\alpha u \in L^p(\Omega)\ \forall |\alpha| \le k \}$$

and we equip it with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \le k} \|\partial^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p}, \quad 1 \le p < \infty$$

and

$$\|u\|_{W^{k,\infty}(\Omega)} := \max_{|\alpha|\leq k} \|\partial^\alpha u\|_{L^\infty(\Omega)}.$$

The special case $p = 2$ provides us with a Hilbert space $H^k(\Omega) := W^{k,2}(\Omega)$ together with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha|\leq k} \langle \partial^\alpha u,\, \partial^\alpha v\rangle_{L^2(\Omega)}$$

For the following purpose, we will have to consider the the space

$$H^1(\Omega) = \left\{u \in L^2(\Omega) \;\middle|\; \nabla u \in L^2(\Omega; \mathbb{R}^d)\right\}$$

together with the induced norm

$$\|u\|_{H^1(\Omega)} = \left(\|u\|^2_{L^2(\Omega)} + \|\nabla u\|_{L^2(\Omega;R^d)}\right)^{1/2}$$

where the second L^2–term is understood as

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^d)} = \left(\int_\Omega |\nabla u(x)|^2 \,\mathrm{d}x\right)^{1/2}.$$

Theorem 2.1 (Trace Theorem)**.**

3 Signorini's Problem

3.1 Physical and Geometric Setup of the Problem

We consider an elastic body occupying a reference configuration $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$. Under the action of external forces and possible contact with a rigid foundation, the body deforms. In the *small–strain* framework, the deformation is described by the *displacement field* $u : \Omega \to \mathbb{R}^d$, and a material point $x \in \Omega$ is mapped to its deformed position

$$\varphi(x) := x' := x + u(x), \quad x \in \overline{\Omega}.$$

The aim of Signorini's problem is to determine an *equilibrium* displacement u under standard boundary conditions and *unilateral* contact constraints on the part of the boundary that may touch the rigid foundation. Throughout the whole report, $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$ denotes an open, bounded and connected *Lipschitz* domain. The later means, that the boundary $\Gamma := \partial\Omega$ can locally be described as the graph of a Lipschitz continuous function. This ensures for example, that the outward normal vector $\mathrm{n}(x)$ is well–defined almost everywhere. The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \mathbin{\dot\cup} \overline{\Gamma_N} \mathbin{\dot\cup} \overline{\Gamma_C}$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions to model different physical interactions (See Figure 1). On Γ_D, we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D,

$$\forall x \in \Gamma_D : \; x + u(x) = x \quad \Longrightarrow \quad \mathrm{Tr}(u)|_{\Gamma_D} = 0.$$

The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. We assume that Γ_C and Γ_D have positive $(d-1)$–Lebesgue measure. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C. While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\mathrm{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u. Later on, the boundary conditions imposed on Γ_C enforce *unilateral* contact through *nonpenetration* and *complementarity*.

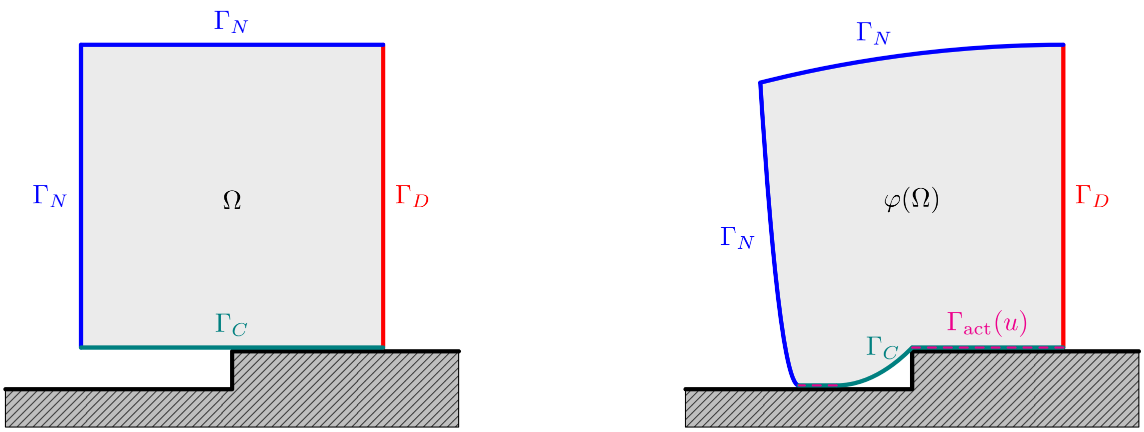

Figure 1: 2D example of Signorini's problem before (left) and after displacement (right).

3.2 Weak Formulation

3.2.1 Admissible Displacements

We start by introducing the energy space and the admissible set of displacements. Homogeneous Dirichlet boundary conditions on Γ_D are incorporated by

$$V := \left\{ v \in H^1(\Omega;\mathbb{R}^d) \;\middle|\; \mathrm{Tr}(v) = 0 \text{ on } \Gamma_D \right\} \subset H^1(\Omega;\mathbb{R}^d). \tag{3.1}$$

Let $\gamma = \mathrm{Tr} : H^1(\Omega;\mathbb{R}^d) \to H^{1/2}(\Gamma;\mathbb{R}^d)$ denote the trace operator. For $v \in V$ we define the normal component of the boundary displacement on Γ_C by

$$v_{\mathrm{n}} := (\gamma v) \cdot \mathrm{n} \quad \text{on } \Gamma_C,$$

where n is the outward unit normal on Γ. The unilateral nature of contact is expressed by the *nonpenetration* constraint. The rigid foundation is located on the exterior side of Γ_C in direction of n. Hence the body is not allowed to move across Γ_C into the obstacle. In the small–strain setting this yields

$$u_{\mathrm{n}} \leq 0 \quad \text{a.e. on } \Gamma_C.$$

This motivates the nonempty, closed and convex cone ([1, Prop. 3.1]) of admissible displacements

$$K := \{ v \in V \mid v_{\mathrm{n}} \leq 0 \text{ a.e. on } \Gamma_C \} \subset V. \tag{3.2}$$

3.2.2 Small Strain Elasticity

We start by defining the *small strain tensor* for a displacement function $u \in H^1(\Omega;\mathbb{R}^d)$, which is similar to the symmetric part of square matrices but for the gradient ∇u of u and through which the deformation of the body $\overline{\Omega}$ by u is quantified.

Definition 3.1 (Small Strain Tensor)**.** *The* `small strain tensor` ε *is defined by*

$$\varepsilon(u) := \frac{1}{2}\left(\nabla u + \nabla u^\top\right) \in L^2(\Omega;\mathrm{Sym}^2(\mathbb{R}^d)), \quad u \in H^1(\Omega;\mathbb{R}^d).$$

We denote by $\mathrm{Sym}^2(\mathbb{R}^d)$ the space of symmetric second order tensors,

$$\mathrm{Sym}^2(\mathbb{R}^d) := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \;\middle|\; s^\top = s \right\} \cong \left\{ S \in \mathbb{R}^{d\times d} \;\middle|\; S^\top = S \right\}.$$

For $s, t \in \mathrm{Sym}^2(\mathbb{R}^d)$ we write denote by

$$s : t := \sum_{i=1}^{d} \sum_{j=1}^{d} s_{ij}\, t_{ij}$$

If $A(x)$ is a fourth order tensor and $s \in \mathrm{Sym}^2(\mathbb{R}^d)$, we use the *double contraction*

$$(A(x) : s)_{ij} := \sum_{k=1}^{d} \sum_{l=1}^{d} A_{ijkl}(x) s_{kl}, \quad s : A(x) : s := \sum_{i=1}^{d} \sum_{j=1}^{d} \sum_{k=1}^{d} \sum_{l=1}^{d} A_{ijkl}(x) s_{ij} s_{kl}.$$

In linear elasticity, stresses are assumed to depend linearly on strains. The material response is encoded in the fourth order *elasticity tensor* A, which maps a symmetric strain tensor to the corresponding *Cauchy stress tensor*. The tensor field A is prescribed by the material and may vary in space, describing a *heterogeneous* body.

Definition 3.2 (Elasticity tensor field)**.** *An* `elasticity tensor field` *is a mapping*

$$A : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$$

such that the coefficients satisfy the symmetries

$$A_{ijkl}(x) = A_{jikl}(x) = A_{ijlk}(x) = A_{klij}(x), \quad \forall 1 \le i, j, k, l \le d,$$

and the uniform ellipticity condition, i.e. there exists $\alpha_A > 0$ such that for a.e. $x \in \Omega$

$$s : A(x) : s \ge \alpha_A \,(s : s) \quad \forall s \in \mathrm{Sym}^2(\mathbb{R}^d).$$

Moreover, there exists $C_A > 0$ such that for a.e. $x \in \Omega$

$$|A_{ijkl}(x)| \le C_A \quad \forall 1 \le i, j, k, l \le d.$$

The *internal* and *surface forces* in the body $\overline{\Omega}$ are described by the *Cauchy stress tensor*.

Definition 3.3 (Cauchy stress tensor)**.** *Let A be an elasticity tensor field. For $u \in H^1(\Omega; \mathbb{R}^d)$ we define the* (linear) Cauchy stress tensor *as the field $\sigma(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ given a.e. by*

$$\sigma(u)(x) := A(x) : \varepsilon(u)(x).$$

On V, we define the symmetric bilinear form

$$a(u, v) := \int_\Omega \sigma(u) : \varepsilon(v) \,\mathrm{d}x, \quad u, v \in V \tag{3.3}$$

representing the *virtual work of internal forces* and the linear bounded functional

$$\ell(v) := \int_\Omega f \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma v \,\mathrm{d}s, \quad v \in V, \tag{3.4}$$

with *volume force density* $f \in L^2(\Omega; \mathbb{R}^d)$ on the body and *surface loads* $F \in L^2(\Gamma_N; \mathbb{R}^d)$ on the Neumann boundary, representing the *virtual work of external forces.* In the following, we want to prove standard properties of a and ℓ. For this, we will need the following inequalities. The first inequality is a result of *Korn's second inequality* ([1, Theorem 3.3] and [1, Section 3.5.2]) and the *Peetre–Tartar* Lemma ([1, Lemma 3.5]). Therefore only the proof of the second one is given.

Lemma 3.1. *Assuming $|\Gamma_D| > 0$, there exists a constant $C_K > 0$ such that*

$$C_K^{-1} \|v\|_{1,\Omega} \le \|\varepsilon(v)\|_{0,\Omega} \le \|v\|_{1,\Omega} \quad \forall v \in V.$$

Proof. We prove the second inequality. For all $v \in V$ we have

$$|\varepsilon(v)(x)| = \left|\frac{1}{2}(\nabla v(x) + \nabla v^\top)\right| \le \frac{1}{2}(|\nabla v(x)| + |\nabla v(x)^\top|) = |\nabla v(x)|$$

and thus

$$\|\varepsilon(v)\|_{0,\Omega}^2 = \int_\Omega |\varepsilon(v)(x)|^2 \,\mathrm{d}x \le \int_\Omega |\nabla v(x)|^2 \,\mathrm{d}x = \|\nabla v\|_{0,\Omega}^2 \le \|v\|_{1,\Omega}^2,$$

where the last inequality holds due to the definition

$$\|v\|_{1,\Omega}^2 = \|v\|_{0,\Omega}^2 + \|\nabla v\|_{0,\Omega}^2.$$

■

Proposition 3.1. *The symmetric bilinear form $a : V \times V \to \mathbb{R}$ defined in (3.3) is*

i) bounded, *i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \quad \textit{with} \quad C_1 = C_A,$$

ii) V-elliptic, *i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2 \quad \textit{with} \quad C_2 = \frac{\alpha_A}{C_K^2}.$$

Moreover, the linear functional ℓ defined in (3.4) is bounded, *i.e. for all $v \in V$ there exists $C_3 > 0$ such that*

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \quad \textit{with} \quad C_3 = \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega}. \tag{3.5}$$

Proof. Let $v, w \in V$ be arbitrary. Then

$$|a(v, w)| \leq \|A : \varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega}$$

by the Cauchy–Schwarz inequality. With Definition 3.2 there exists $C_A > 0$ such that

$$\|A : \varepsilon(v)\|_{0,\Omega} \leq C_A \|\varepsilon(v)\|_{0,\Omega}.$$

Combining this with the second inequality form Lemma 3.1 we have

$$|a(v, w)| \leq C_A \|\varepsilon(v)\|_{0,\Omega} \|\varepsilon(w)\|_{0,\Omega} \leq C_A \|v\|_{1,\Omega} \|w\|_{1,\Omega}.$$

The ellipticity of a follows by

$$a(v, v) = \int_\Omega \varepsilon(v) : A : \varepsilon(v) \, \mathrm{d}x \geq \alpha_A \int_\Omega \varepsilon(v) : \varepsilon(v) = \alpha_A \|\varepsilon(v)\|_{0,\Omega}^2,$$

where $\alpha_A > 0$ exists due to Definition 3.2. The first inequality of Lemma 3.1 finally yields

$$a(v, v) \geq \alpha_A \|\varepsilon(v)\|_{0,\Omega}^2 \geq \alpha_A \left(C_K^{-1} \|v\|_{1,\Omega}\right)^2 = \frac{\alpha_A}{C_K^2} \|v\|_{1,\Omega}^2.$$

Again by Cauchy–Schwarz inequality,

$$|\ell(v)| \leq \|f\|_{0,\Omega} \|v\|_{0,\Omega} + \|F\|_{0,\Gamma_N} \|\gamma v\|_{0,\Gamma_N} \quad \forall v \in V.$$

By the Trace Theorem 2.1 there exists $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma v\|_{0,\Gamma_N} \leq \|\gamma v\|_{0,\Gamma} \leq \|\gamma v\|_{\frac{1}{2},\Gamma} \leq C_{\mathrm{tr}} \|v\|_{1,\Omega} \quad \forall v \in V.$$

Finally,

$$|\ell(v)| \leq \left(\|f\|_{0,\Omega} + C_{\mathrm{tr}} \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega} \leq \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega} \quad \forall v \in V.$$

■

3.2.3 Energy Minimization and Well-posedness

The total potential energy of the elastic body is given by the quadratic functional

$$\mathcal{J}(v) := \frac{1}{2} a(v, v) - \ell(v), \quad v \in V. \tag{3.6}$$

In the presence of unilateral contact, the admissible displacements are restricted to the closed convex set $K \subset V$. We therefore consider the following constrained minimization problem.

Problem 3.1 (Signorini's Problem as Energy Minimization)**.** *Find $u \in K$ such that*

$$\mathcal{J}(u) = \min_{v\in K} \mathcal{J}(v) = \min_{v\in K} \frac{1}{2} a(v,v) - \ell(v).$$

We now what to prove well-posedness of the above Problem 3.1. This consist of existence and uniqueness of a solution and continuous dependency of the solution on the load functional ℓ. We start by the uniqueness, which follows directly by the strict convexity of $\mathcal{J}$ and the convexity of K.

Lemma 3.2 (Strict Convexity of $\mathcal{J}$)**.** *The Functional $\mathcal{J}$ defined in* (3.6) *is strictly convex.*

Proof. For $u, v \in V$ and $t \in [0,1]$, we set $w_t := (1-t)u + tv$. Then

$$a(w_t, w_t) = (1-t)a(u,u) + ta(v,v) - t(1-t)a(u-v, u-v)$$

and

$$\ell(w_t) = (1-t)\ell(u) + t\ell(v).$$

Together, we have

$$\mathcal{J}(w_t) = (1-t)\mathcal{J}(u) + t\mathcal{J}(v) - \frac{1}{2}t(1-t)a(u-v,u-v).$$

If $u \neq v$ then $u - v \neq 0$ and thus by V-ellipticity of a we have $a(u-v, u-v) > 0$ and $t(1-t) > 0$. This implies

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u) + t\mathcal{J}(v).$$

■

Proposition 3.2 (Uniqueness of Minimizer of $\mathcal{J}$)**.** *If there exists a minimizer of the functional $\mathcal{J}$ defined in* (3.6)*, then it is unique.*

Proof. Assume that there exist two minimizers $u_1, u_2 \in K$ of $\mathcal{J}$ on K, i.e.

$$\mathcal{J}(u_1) = \min_{v\in K} \mathcal{J}(v) = \mathcal{J}(u_2). \tag{3.7}$$

If $u_1 = u_2$ there is nothing to show. Hence assume $u_1 \neq u_2$. Since K is convex, for every $t \in (0,1)$ the convex combination

$$w_t := (1-t)u_1 + tu_2$$

belongs to K. By Lemma 3.2, $\mathcal{J}$ is strictly convex and therefore we have

$$\mathcal{J}(w_t) < (1-t)\mathcal{J}(u_1) + t\mathcal{J}(u_2) \quad \forall t \in (0,1).$$

Using (3.7), this yields

$$\mathcal{J}(w_t) < (1-t)\min_{v\in K} \mathcal{J}(v) + t \min_{v\in K} \mathcal{J}(v) = \min_{v\in K} \mathcal{J}(v).$$

This contradicts the definition of the minimum, since $w_t \in K$. Therefore the assumption $u_1 \neq u_2$ is false and we conclude $u_1 = u_2$. ■

It remains to show the existence of at least one minimizer. For this, we verify *coercivity* and *weakly lower semicontinuity* of the functional $\mathcal{J}$. The proof requires the following lemma.

Lemma 3.3 (Weak lower semicontinuity of the norm). *Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \to \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X, then*

$$\|x\|_X \leq \liminf_{j\to\infty} \|x_j\|_X.$$

Proof. Let $J : X \to X^{**}$ be the canonical embedding defined by

$$(Jx)(\varphi) := \varphi(x) \quad \forall \varphi \in X^*.$$

By the Hahn–Banach theorem, J is an isometry, i.e. $\|Jx\|_{X^{**}} = \|x\|_X$ for all $x \in X$. Therefore,

$$\|x\|_X = \|Jx\|_{X^{**}} = \sup_{\substack{\varphi\in X^* \\ \|\varphi\|_{X^*}\leq 1}} |(Jx)(\varphi)| = \sup_{\substack{\varphi\in X^* \\ \|\varphi\|_{X^*}\leq 1}} |\varphi(x)|.$$

If $x_j \rightharpoonup x$ in X, then $\varphi(x_j) \to \varphi(x)$ for all $\varphi \in X^*$ and hence

$$\|x\|_X = \sup_{\substack{\varphi\in X^* \\ \|\varphi\|_{X^*}\leq 1}} |\varphi(x)| \leq \liminf_{j\to\infty} \sup_{\substack{\varphi\in X^* \\ \|\varphi\|_{X^*}\leq 1}} |\varphi(x_j)| = \liminf_{j\to\infty} \|x_j\|_X.$$

■

Lemma 3.4 ($\mathcal{J}$ is coercive and w.l.s.c.). *The functional $\mathcal{J}$ defined in (3.6) is*

i) coercive *on V, i.e.*

$$\mathcal{J}(v) \to \infty \quad \text{as} \quad \|v\|_{1,\Omega} \to \infty,$$

ii) weakly lower semicontinuous *on V, i.e. if $v_j \rightharpoonup v$ in V, then*

$$\mathcal{J}(v) \leq \liminf_{j\to\infty} \mathcal{J}(v_j).$$

Proof. By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{-1,\Omega}\|v\|_{1,\Omega}$. Hence

$$\mathcal{J}(v) = \frac{1}{2}\, a(v,v) - \ell(v) \geq \frac{\alpha}{2}\, \|v\|_{1,\Omega}^2 - \|\ell\|_{-1,\Omega}\, \|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega}\to\infty} \infty.$$

For $s \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ set

$$\|s\|_A^2 := \int_\Omega s : A : s \,\mathrm{d}x.$$

By uniform ellipticity of A, the map $\|\cdot\|_A$ defines a norm on $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ equivalent to $\|\cdot\|_{0,\Omega}$. Moreover, $a(v,v) = \|\varepsilon(v)\|_A^2$. The mapping $v \mapsto \varepsilon(v)$ is linear and bounded from V to $L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$. Hence, if $v_j \rightharpoonup v$ in V, then

$$\varepsilon(v_j) \rightharpoonup \varepsilon(v) \quad \text{in } L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)).$$

Since norms are weakly lower semicontinuous by Lemma 3.3, we obtain

$$a(v,v) = \|\varepsilon(v)\|_A^2 \leq \liminf_{j\to\infty} \|\varepsilon(v_j)\|_A^2 = \liminf_{j\to\infty} a(v_j, v_j).$$

Furthermore, $\ell \in V'$ is weakly continuous, hence $\ell(v_j) \to \ell(v)$. Therefore,

$$\mathcal{J}(v) = \frac{1}{2}\, a(v,v) - \ell(v) \leq \frac{1}{2}\, \liminf_{j\to\infty} a(v_j, v_j) - \lim_{j\to\infty} \ell(v_j) = \liminf_{j\to\infty} \mathcal{J}(v_j).$$

■

Proposition 3.3 (Existence of Minimizer of $\mathcal{J}$)**.** *The functional $\mathcal{J}$ defined in 3.6 has at least one minimizer $u \in K$.*

Proof. Set

$$M := \inf_{v \in K} \mathcal{J}(v) \in [-\infty, \infty).$$

Since $0 \in K$ we have $M \leq \mathcal{J}(0) = 0$, hence M is finite from above. Let $\{v_j\}_{j \in \mathbb{N}} \subset K$ be a minimizing sequence, i.e. $\mathcal{J}(v_j) \to M$ as $j \to \infty$. By Lemma 3.4 *i)*, $\mathcal{J}$ is coercive and thus, the sequence $\{v_j\}_{j \in \mathbb{N}}$ is bounded in V. Since V is a Hilbert space, there exist a subsequence $\{v_{j_k}\}_{k \in \mathbb{N}}$ and some $u \in V$ such that

$$v_{j_k} \rightharpoonup u \quad \text{in } V.$$

As $K \subset V$ is closed and convex, it is weakly closed in V. Hence $u \in K$. By Lemma 3.4 *ii)*, $\mathcal{J}$ is weakly lower semicontinuity and thus we obtain

$$\mathcal{J}(u) \leq \liminf_{k \to \infty} \mathcal{J}(v_{j_k}) = M = \inf_{v \in K} \mathcal{J}(v),$$

so u is a minimizer of $\mathcal{J}$ on K. ∎

Corollary 3.1 (Well-posedness of Problem 3.1)**.** *The functional $\mathcal{J}$ defined in 3.6 admits a unique minimizer $u \in K$. Moreover, the minimizer satisfies the a-priori bound*

$$\|u\|_{1,\Omega} \leq \frac{2\,C_K^2 \max\{1, C_{\mathrm{tr}}\}}{\alpha_A} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right). \tag{3.8}$$

Consequently, Problem 3.1 has a unique stable solution.

Proof. By Proposition 3.3 we have existence of a solution $u \in K$ to Problem 3.1 and by Proposition 3.2 it is unique. It remains to prove the a-priori bound. Since u is the unique minimizer, it holds

$$\mathcal{J}(u) \leq \mathcal{J}(0) = 0 \quad \Longrightarrow \quad \frac{1}{2}\, a(u,u) \leq \ell(u).$$

By Proposition 3.1 *ii)* we have

$$a(u,u) \geq \frac{\alpha_A}{C_K^2}\, \|u\|_{1,\Omega}^2$$

and thus

$$\frac{\alpha_A}{2\,C_K^2}\, \|u\|_{1,\Omega}^2 \leq \frac{1}{2}\, a(u,u) \leq |\ell(u)| \leq \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right) \|u\|_{1,\Omega},$$

where the last inequality is (3.5). Dividing both sides by $\|u\|_{1,\Omega}$ yields the desired estimate (3.8). For $u = 0$ the estimate is trivial. ∎

3.2.4 Variational inequality as an optimality condition

Finally, we turn to the *weak formulation* of Signorini's problem. Unlike the weak formulation of conventional simple boundary value problems, here we do not end up with a variational equation, but with a variational inequality. This is derived by a first-order optimality condition on the functional $\mathcal{J}$.

Lemma 3.5 (Fréchet differentiability of $\mathcal{J}$)**.** *The functional $\mathcal{J}$ defined in (3.6) is Fréchet differentiable on V. Its derivative at $u \in V$ is given by*

$$\mathcal{J}'(u)[h] = a(u,h) - \ell(h) \quad \forall h \in V. \tag{3.9}$$

Proof. Let $u, h \in V$ and $t \in \mathbb{R}$. Using bilinearity of a and linearity of ℓ we compute

$$\mathcal{J}(u+th) - \mathcal{J}(u) = \frac{1}{2}\left(a(u+th, u+th) - a(u,u)\right) - t\ell(h) = t\left(a(u,h) - \ell(h)\right) + \frac{t^2}{2}a(h,h).$$

Hence

$$\frac{\mathcal{J}(u+th) - \mathcal{J}(u) - t(a(u,h) - \ell(h))}{\|th\|_{1,\Omega}} = \frac{|t|}{2}\frac{|a(h,h)|}{\|h\|_{1,\Omega}} \leq \frac{|t|}{2} C_A \|h\|_{1,\Omega} \xrightarrow{t\to 0} 0,$$

by boundedness of a as proven in Proposition 3.1. ∎

Proposition 3.4 (First-order optimality condition on convex sets)**.** *A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality*

$$\mathcal{J}'(u)[v-u] \geq 0 \quad \forall v \in K. \tag{3.10}$$

Proof. Assume first that $u \in K$ minimizes $\mathcal{J}$ over K and let $v \in K$. By convexity of K we have $u + t(v-u) \in K$ for all $t \in [0,1]$. Define $\phi(t) := \mathcal{J}(u + t(v-u))$. Then ϕ attains its minimum at $t = 0$, hence $\phi'(0^+) \geq 0$. Using the chain rule and Fréchet differentiability of $\mathcal{J}$ we obtain

$$\phi'(0^+) = \mathcal{J}'(u)[v-u] \geq 0.$$

Conversely, assume (3.10). Let $v \in K$ be arbitrary and set $h := v - u$. By Taylor expansion of the quadratic functional $\mathcal{J}$ we have

$$\mathcal{J}(u+h) - \mathcal{J}(u) = \mathcal{J}'(u)[h] + \frac{1}{2}a(h,h).$$

By (3.10), $\mathcal{J}'(u)[h] \geq 0$, and since $a(h,h) \geq 0$ we conclude $\mathcal{J}(v) \geq \mathcal{J}(u)$ for all $v \in K$. Hence u is a minimizer over K. ∎

Combining Lemma 3.5 and Proposition 3.4, $u \in K$ solves Problem 3.1 if and only if

$$\mathcal{J}'(u)[v-u] = a(u, v-u) - \ell(v-u) \geq 0 \quad \forall v \in K.$$

This is the first weak formulation of Signorini's problem as a variational inequality.

Problem 3.2 (Weak Formulation of Signorini's Problem)**.** *Find $u \in K$ such that*

$$a(u, v-u) \geq \ell(v-u) \quad \forall v \in K.$$

Well-posedness of Problem 3.2 is already ensured, since it is equivalent to the minimization Problem 3.1 in the sense that each solution to the one of the problems is a solution to the other problem (Proposition 3.4). The solution is unique due to Corollary 3.1 with the stability estimate (3.8). Another useful equivalent formulation is the following.

Problem 3.3 (Weak Formulation of Signorini's Problem)**.** *Find $u \in K$ such that*

$$\begin{cases} a(u,u) = \ell(u), \\ a(u,v) \geq \ell(v) \quad \forall v \in K. \end{cases}$$

Proposition 3.5 (Equivalence of Problem 3.2 and Problem 3.3)**.** *Any solution $u \in K$ to Problem 3.2 is also a solution to Problem 3.3 and vise versa.*

Proof. Let $u \in K$ solve Problem 3.3. Then

$$a(u, v-u) = a(u,v) - a(u,u) = a(u,v) - \ell(u) \geq \ell(v) - \ell(u) = \ell(v-u).$$

Assuming that $u \in K$ solves Problem 3.2 and testing $v = 0$

$$a(u,-u) \geq \ell(-u) \quad \iff \quad -a(u,u) \geq -\ell(u) \quad \iff \quad a(u,u) \leq \ell(u).$$

However, testing with $v = 2u$ yields $a(u,u) \geq \ell(u)$ and thus $a(u,u) = \ell(u)$. Furthermore, Problem 3.2 reads

$$a(u, v-u) = a(u,v) - a(u,u) \geq \ell(v-u) = \ell(v) - \ell(u) \quad \Longrightarrow \quad a(u,v) \geq a(u,u) + \ell(v) - \ell(u).$$

Plugging in $a(u,u) = \ell(u)$ yields $a(u,v) \geq \ell(v)$. ■

We close this section by providing another stability result on the dependency of solutions to varying source terms as input data.

Proposition 3.6 (Lipschitz dependency)**.** *Let $f_1, f_2 \in L^2(\Omega;\mathbb{R}^d)$ and $F_1, F_2 \in L^2(\Gamma_N;\mathbb{R}^d)$ and let $\ell_i \in V'$ be the corresponding load functionals*

$$\ell_i(v) = \int_\Omega f_i \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F_i \cdot \gamma v \,\mathrm{d}s, \quad v \in V, \;\; i = 1,2.$$

Let $u_i \in K$ be the corresponding (unique) minimizers of

$$\mathcal{J}_i(v) := \frac{1}{2}\, a(v,v) - \ell_i(v)$$

over K. Then the solutions depend Lipschitz continuously on the data, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \leq \frac{C_K^2}{\alpha_A}\, \|\ell_1 - \ell_2\|_{-1,\Omega} \leq \frac{C_K^2 \,\max\{1, C_{\mathrm{tr}}\}}{\alpha_A} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\right).$$

Proof. Since $u_i \in K$ minimizes $\mathcal{J}_i$ over the closed convex set K by Proposition 3.4, it satisfies the first-order optimality condition

$$a(u_i, v - u_i) \geq \ell_i(v - u_i) \quad \forall v \in K, \;\; i = 1,2. \tag{3.11}$$

Choose $v = u_2$ in (3.11) for $i = 1$ and $v = u_1$ in (3.11) for $i = 2$. This yields

$$\begin{aligned} a(u_1, u_2 - u_1) &\geq \ell_1(u_2 - u_1), \\ a(u_2, u_1 - u_2) &\geq \ell_2(u_1 - u_2). \end{aligned}$$

Adding both inequalities and using bilinearity of a gives

$$a(u_1 - u_2, u_1 - u_2) \leq (\ell_1 - \ell_2)(u_1 - u_2).$$

By the V-ellipticity estimate from Proposition 3.1 ii) we have

$$a(w,w) \geq \frac{\alpha_A}{C_K^2}\, \|w\|_{1,\Omega}^2 \quad \forall w \in V.$$

Applying this with $w = u_1 - u_2$ and estimating the right-hand side by the dual norm $\|\cdot\|_{-1,\Omega}$ yields

$$\frac{\alpha_A}{C_K^2}\, \|u_1 - u_2\|_{1,\Omega}^2 \leq a(u_1 - u_2, u_1 - u_2) \leq \|\ell_1 - \ell_2\|_{-1,\Omega}\, \|u_1 - u_2\|_{1,\Omega}.$$

If $u_1 = u_2$ there is nothing to show. Otherwise we divide by $\|u_1 - u_2\|_{1,\Omega}$ and obtain

$$\|u_1 - u_2\|_{1,\Omega} \le \frac{C_K^2}{\alpha_A} \|\ell_1 - \ell_2\|_{-1,\Omega}. \tag{3.12}$$

It remains to bound $\|\ell_1 - \ell_2\|_{-1,\Omega}$ in terms of $f_1 - f_2$ and $F_1 - F_2$. Let $v \in V$ be arbitrary. By Cauchy–Schwarz,

$$|(\ell_1 - \ell_2)(v)| \le \|f_1 - f_2\|_{0,\Omega} \|v\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \|\gamma v\|_{0,\Gamma_N}.$$

Using $\|v\|_{0,\Omega} \le \|v\|_{1,\Omega}$ and the trace estimate $\|\gamma v\|_{0,\Gamma_N} \le C_{\mathrm{tr}} \|v\|_{1,\Omega}$ we obtain

$$|(\ell_1 - \ell_2)(v)| \le \max\{1, C_{\mathrm{tr}}\} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\right) \|v\|_{1,\Omega}.$$

Dividing by $\|v\|_{1,\Omega}$ and taking the supremum over $0 \ne v \in V$ yields

$$\|\ell_1 - \ell_2\|_{-1,\Omega} \le \max\{1, C_{\mathrm{tr}}\} \left(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N}\right)$$

Combining this with (3.12) gives the claim. ■

3.3 Strong Formulation

From the previous weak formulations of Signorini's problem, we know there exists a uniquely determined $u \in K$ that solves the problem. To derive a *strong* PDE formulation, we have to discuss what was $\sigma(u)\mathrm{n}$ means and in what sense the contact conditions on Γ_C can be described as equalities and inequalities.

3.3.1 Additional regularity and meaning of boundary tractions

In the weak setting, the solution only satisfies $u \in K \subset H^1(\Omega; \mathbb{R}^d)$., i.e. $\varepsilon(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ and, by $\sigma(u) = A : \varepsilon(u)$, we obtain $\sigma(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$. This regularity is sufficient to interpret the internal virtual work

$$a(u, v) = \int_\Omega \sigma(u) : \varepsilon(v)\, \mathrm{d}x,$$

but it is not enough to write the equilibrium equation $-\operatorname{div} \sigma(u) = f$ as an identity in $L^2(\Omega; \mathbb{R}^d)$, since the divergence of an L^2–field is only defined in the sense of distributions. To formulate a strong equilibrium equation in $L^2(\Omega; \mathbb{R}^d)$ and to give a meaning to *boundary tractions*, we assume additional regularity of the stress,

$$\sigma(u) \in H(\mathrm{div}; \Omega) := \left\{ \tau \in L^2(\Omega; \mathbb{R}^{d \times d}) \;\middle|\; \operatorname{div} \tau \in L^2(\Omega; \mathbb{R}^d) \right\}. \tag{3.13}$$

Here $\operatorname{div} \tau$ is taken row-wise, i.e.

$$(\operatorname{div} \tau)_i = \sum_{j=1}^d \partial_j \tau_{ij}$$

in the distributional sense. If $\sigma(u) \in H(\mathrm{div}; \Omega)$, then $\operatorname{div} \sigma(u) \in L^2(\Omega; \mathbb{R}^d)$, so the equilibrium equation is well–defined. The same regularity is also exactly what is needed to define the boundary traction (or surface force density) associated with τ. Formally, for smooth τ one would write

$$\tau\, \mathrm{n} \quad \text{on } \Gamma,$$

where $\mathrm{n} : \Gamma \to \mathbb{R}^d$ is the outward unit normal and $\tau\, \mathrm{n}$ denotes the pointwise matrix–vector product. However, for $\tau \in L^2(\Omega; \mathbb{R}^{d \times d})$ a pointwise boundary trace is not available, so $\tau\, \mathrm{n}$ cannot be defined as an $L^2(\Gamma)$–function in general. One therefore starts from the classical integration by parts formula. If τ and v are smooth, then

$$\int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x = \int_\Gamma (\tau\,\mathrm{n})\cdot v \,\mathrm{d}s. \tag{3.14}$$

The key point now is that the left-hand side still makes sense for $\tau \in H(\mathrm{div};\Omega)$ and $v \in H^1(\Omega;\mathbb{R}^d)$. Hence, for fixed $\tau \in H(\mathrm{div};\Omega)$ the map

$$F_\tau(v) := \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x, \quad v \in H^1(\Omega;\mathbb{R}^d),$$

defines a continuous linear functional on $H^1(\Omega;\mathbb{R}^d)$. The crucial observation is that F_τ depends only on the boundary trace γv. Indeed, if $w \in H^1_0(\Omega;\mathbb{R}^d)$, then $\gamma w = 0$, and we claim that $F_\tau(w) = 0$. To see this, we take a sequence $(w_m)_{m\in\mathbb{N}} \subset C_c^\infty(\Omega;\mathbb{R}^d)$ with $w_m \to w$ in $H^1(\Omega;\mathbb{R}^d)$. Since $\operatorname{div}\tau \in L^2(\Omega;\mathbb{R}^d)$, we have for all $w_m \in C_c^\infty(\Omega;\mathbb{R}^d)$, that

$$\int_\Omega (\operatorname{div}\tau)\cdot w_m \,\mathrm{d}x = -\int_\Omega \tau : \nabla w_m \,\mathrm{d}x,$$

and therefore $F_\tau(w_m) = 0$ for all $m \in \mathbb{N}$. Passing to the limit $m \to \infty$ using the continuity of F_τ on $H^1(\Omega;\mathbb{R}^d)$ yields $F_\tau(w) = 0$. Consequently, if $v_1, v_2 \in H^1(\Omega;\mathbb{R}^d)$ satisfy $\gamma v_1 = \gamma v_2$, then $v_1 - v_2 \in H^1_0(\Omega;\mathbb{R}^d)$ and thus $F_\tau(v_1) = F_\tau(v_2)$. This is exactly the statement that F_τ depends only on γv. Since $\gamma : H^1(\Omega;\mathbb{R}^d) \to H^{1/2}(\Gamma;\mathbb{R}^d)$ is continuous and $\ker(\gamma) = H^1_0(\Omega;\mathbb{R}^d)$, the previous argument implies that F_τ induces a well-defined continuous linear functional on the trace space $H^{1/2}(\Gamma;\mathbb{R}^d)$. Hence there exists a unique element $\gamma_\mathrm{n}(\tau) \in H^{-1/2}(\Gamma;\mathbb{R}^d)$ such that

$$\langle \gamma_\mathrm{n}(\tau), \gamma v\rangle_\Gamma = \int_\Omega \tau : \nabla v \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x \quad \forall v \in H^1(\Omega;\mathbb{R}^d), \tag{3.15}$$

where $\langle\cdot,\cdot\rangle_\Gamma$ denotes the duality pairing of $H^{-1/2}(\Gamma;\mathbb{R}^d) \times H^{1/2}(\Gamma;\mathbb{R}^d)$. The operator $\gamma_\mathrm{n} : H(\mathrm{div};\Omega) \to H^{-1/2}(\Gamma;\mathbb{R}^d)$ is called the *normal trace operator* (see [1, Lemma 3.1]). In the smooth case, (3.15) reduces to (3.14), so $\gamma_\mathrm{n}(\tau)$ coincides with the classical boundary function $\tau\,\mathrm{n}$. For this reason we write suggestively

$$\gamma_\mathrm{n}(\tau) = \tau\,\mathrm{n} \in H^{-1/2}(\Gamma;\mathbb{R}^d).$$

Physically, $\tau\,\mathrm{n}$ is the traction vector (or surface force density) exerted across the boundary with outward normal n, and in particular $t(u) := \sigma(u)\mathrm{n}$ is the surface force induced by the stress field $\sigma(u)$. Finally, since τ is symmetric, $\tau : \nabla v = \tau : \varepsilon(v)$ for all v, so (3.15) can be written in the symmetric form

$$\langle \tau\,\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Omega \tau : \varepsilon(v) \,\mathrm{d}x + \int_\Omega (\operatorname{div}\tau)\cdot v \,\mathrm{d}x \quad \forall (v,\tau) \in H^1(\Omega;\mathbb{R}^d) \times H(\mathrm{div};\Omega). \tag{3.16}$$

Applying this to $\tau = \sigma(u)$ and assuming $\sigma(u) \in H(\mathrm{div};\Omega)$ provides

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v) \,\mathrm{d}x + \int_\Omega (\operatorname{div}\sigma(u))\cdot v \,\mathrm{d}x \quad \forall v \in H^1(\Omega;\mathbb{R}^d).$$

which will be needed in the strong Signorini conditions.

3.4 Mixed Formulation

3.4.1 Mixed Weak Formulation as a Sattlepoint Problem

3.4.2 Well–posedness and inf–sup condition

4 Outlook

A References

[1] Franz Chouly, Patrick Hild, and Yves Renard. *Finite Element Approximation of Contact and Friction in Elasticity*, volume 48 of *Advances in Continuum Mechanics*. Springer International Publishing, 2023.

B List of Figures